

Descartes' picture of pain

De l'homme (Descartes, 1644) 'If for example fire comes near the foot, minute particles of this fire, which you know move at great velocity, have the power to set in motion the spot of skin on the foot which they touch, and by this means pulling on the delicate thread which is attached to the spot of the skin, they open up at the same instant the pore against which the delicate thread ends, just as by pulling on one end of a rope one makes to strike at the same instant a bell which hangs at the end.'



Pain perception and nociception

- The sensations we call **pain** are the most distinctive of all the sensory modalities
- Pain is a submodality of somatic sensation and serves an important protective function: It **warns of injury** that should be avoided or treated
- Pain has an **urgent and primitive** quality, a quality responsible for the **affective and emotional aspect** of pain perception. Moreover, the intensity with which pain is felt is affected by surrounding conditions, and the same stimulus can produce different responses in different individuals under similar conditions

Pain perception and nociception

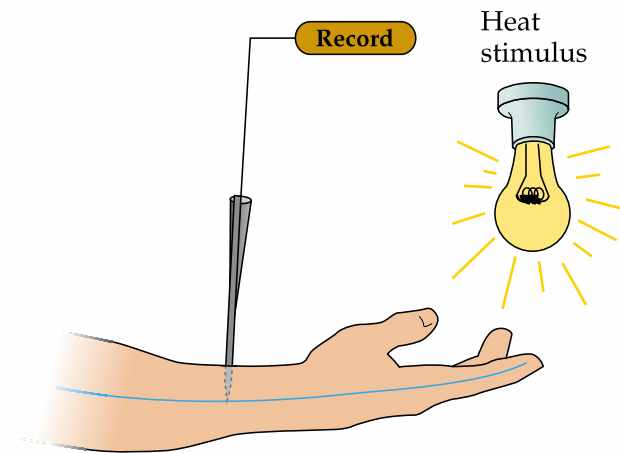
- Pain is a percept; it is an unpleasant sensory and emotional experience associated with actual or potential tissue damage
- Although pain is mediated by the nervous system, a distinction between pain and the neural mechanisms of nociception - the response to perceived or actual tissue damage - is important both clinically and experimentally
- Certain tissues have specialized sensory receptors, called *nociceptors*, that are activated by noxious insults to peripheral tissues
- Nociception does not necessarily lead to the experience of pain

Pain perception and nociception

- There are no “painful stimuli”—stimuli that invariably elicit the perception of pain in all individuals
- Nociceptive pain vs. neuropathic pain

Noxious Insults Activate Nociceptors

(A)

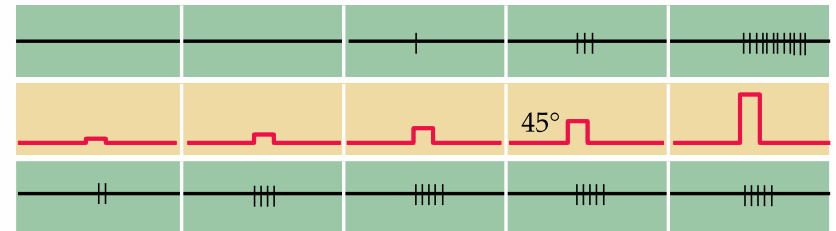


(B)

Nociceptor

Stimulus

Nonnociceptive thermoreceptor



(C)

Magnitude of afferent response (action potentials per second)

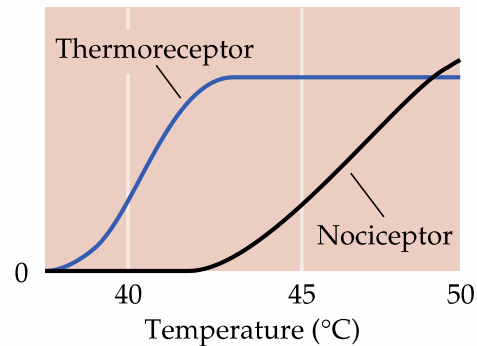
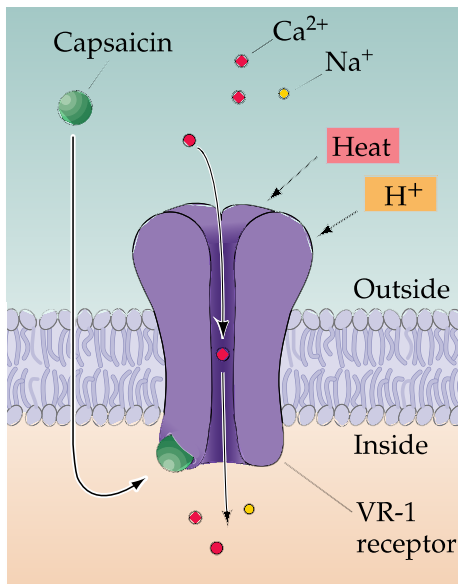
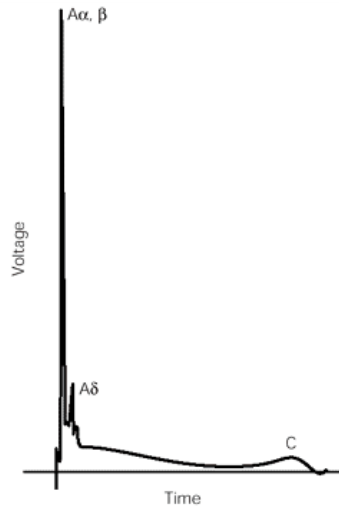


Figure 9.1 Experimental demonstration that nociception involves specialized neurons, not simply greater dis-

Noxious Insults Activate Nociceptors

- *Thermal* nociceptors are activated by extreme temperatures ($>45^{\circ}\text{C}$ or $< 5^{\circ}\text{C}$), small-diameter, thinly myelinated $\text{A}\delta$ fibers that conduct signals at about 5-30 m/s.
- *Mechanical* nociceptors are activated by intensive pressure applied to the skin. They also have thinly myelinated $\text{A}\delta$ fibers conducting at 5-30 m/s
- *Polymodal* nociceptors are activated by high-intensity mechanical, chemical, or thermal (both hot and cold) stimuli. These nociceptors have small-diameter, nonmyelinated C fibers that conduct slowly, generally at velocities of less than 1.0 m/s

Noxious Insults Activate Nociceptors



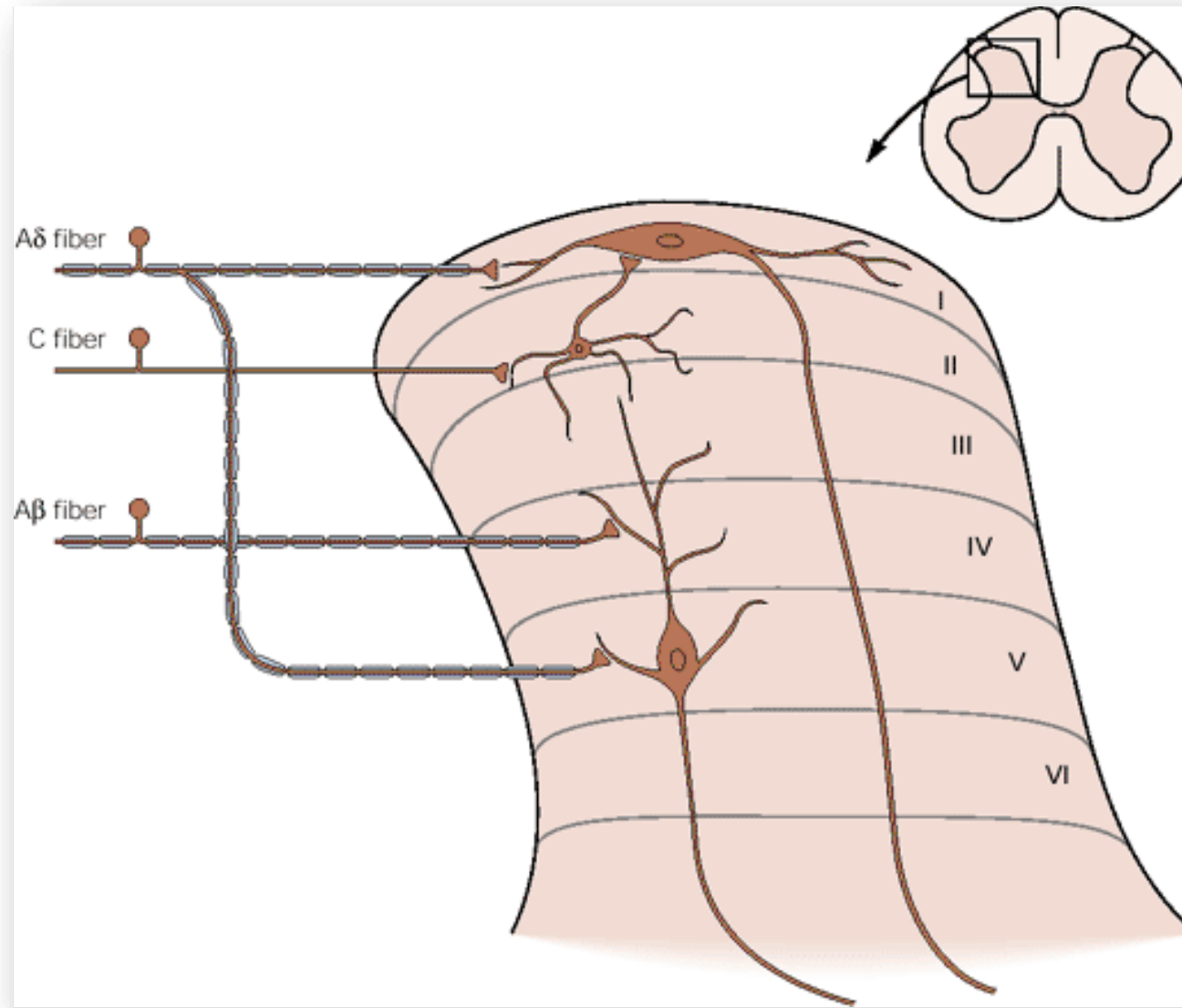
A

A

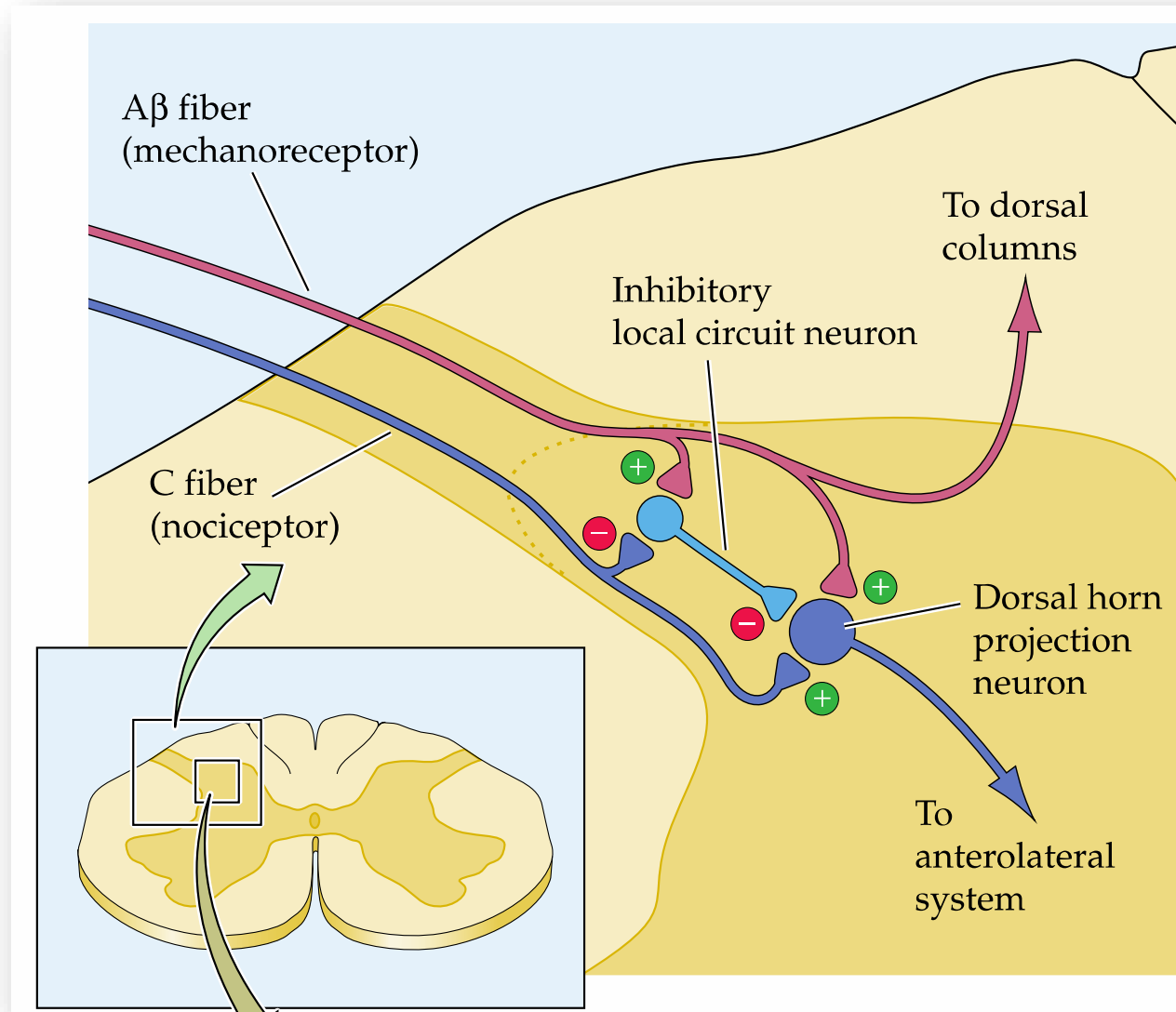
Aα, β

Aα, β

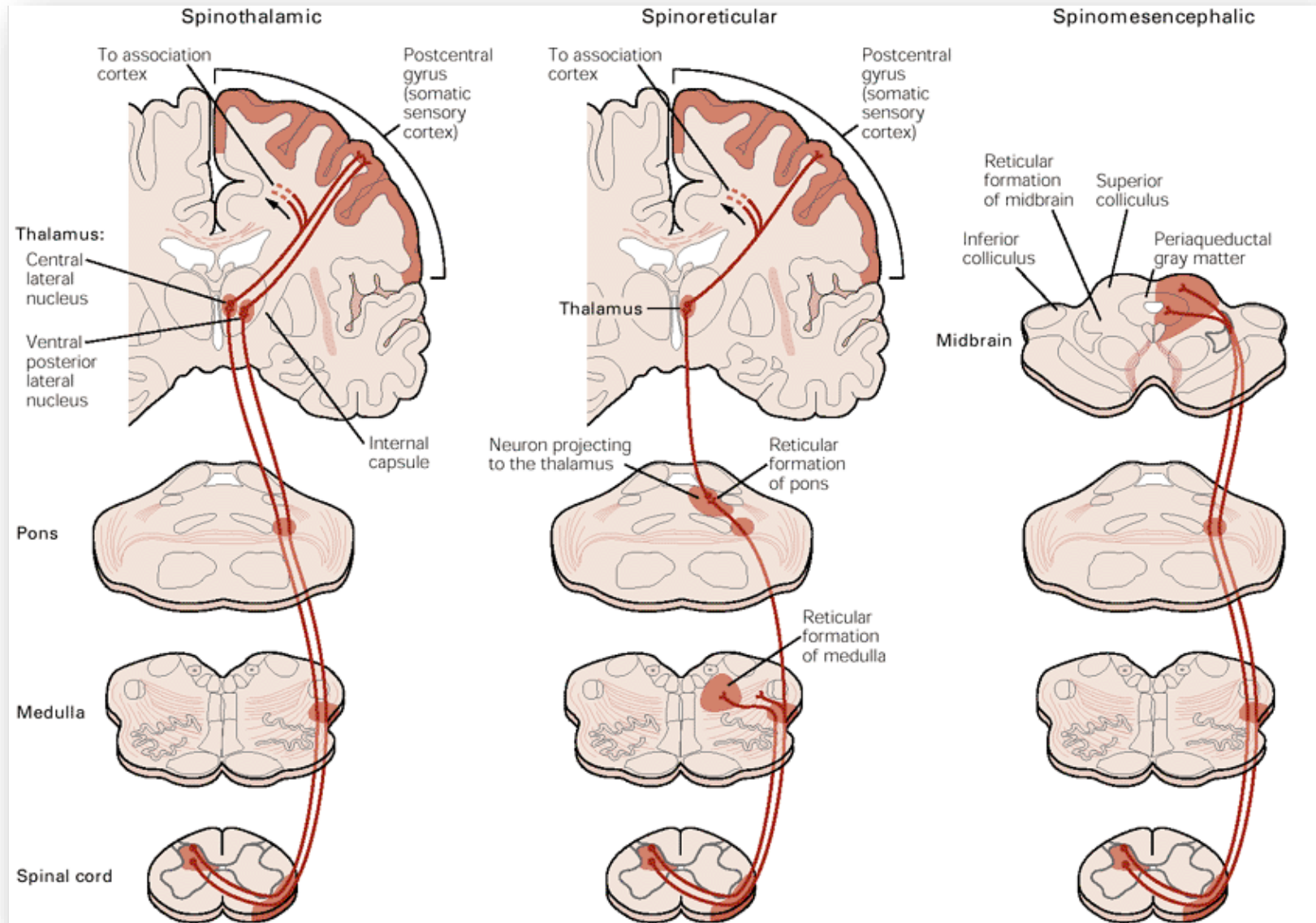
Noxious Insults Activate Nociceptors



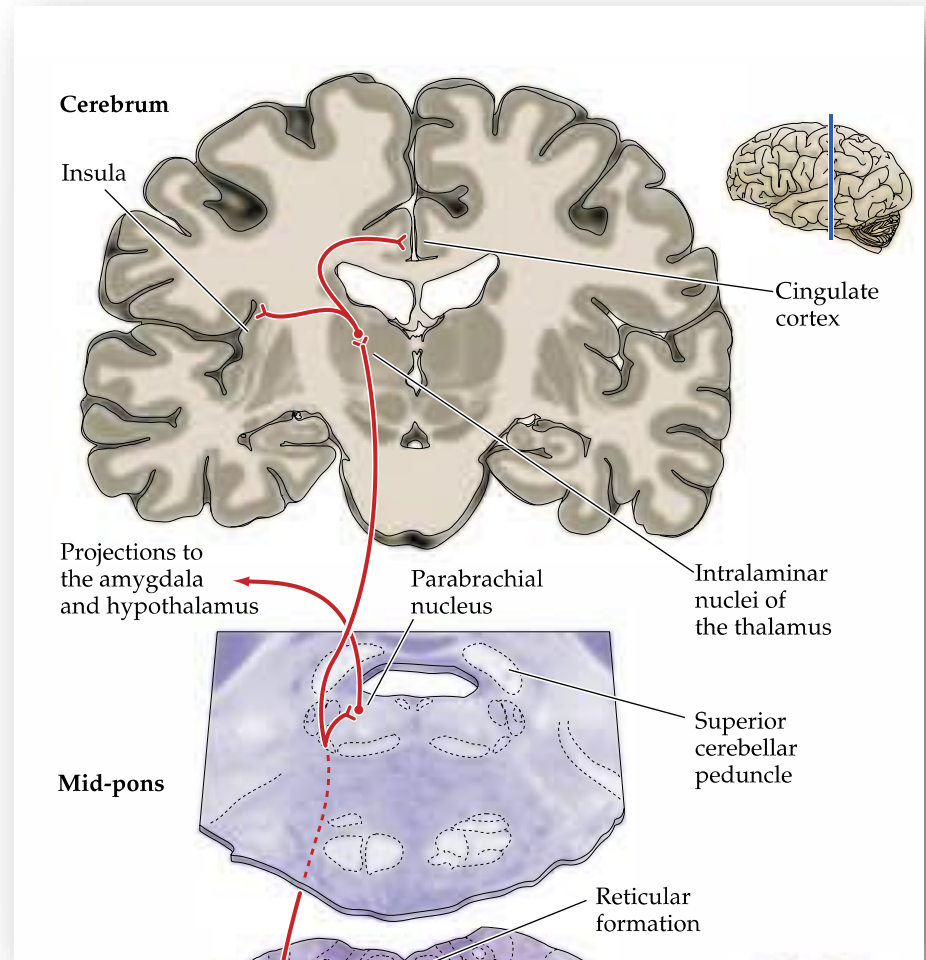
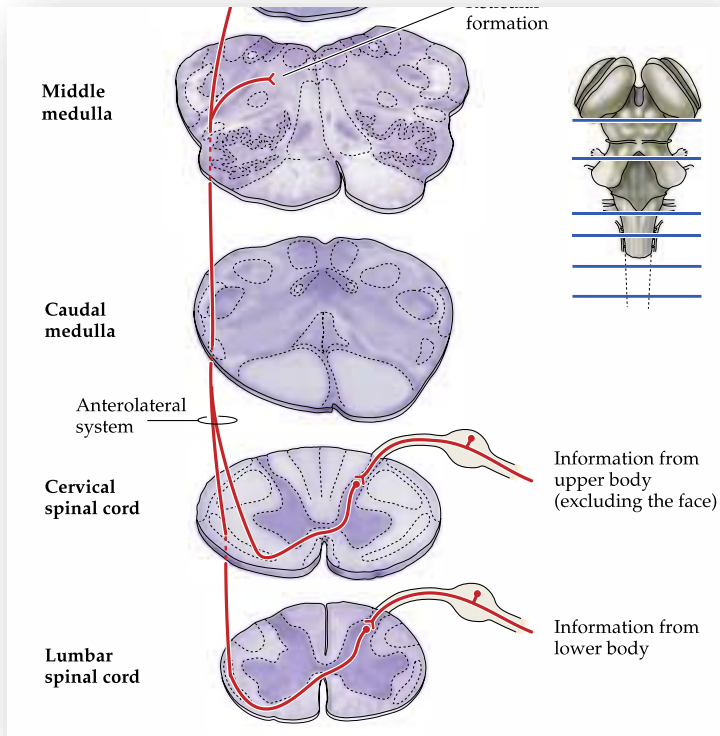
Noxious Insults Activate Nociceptors



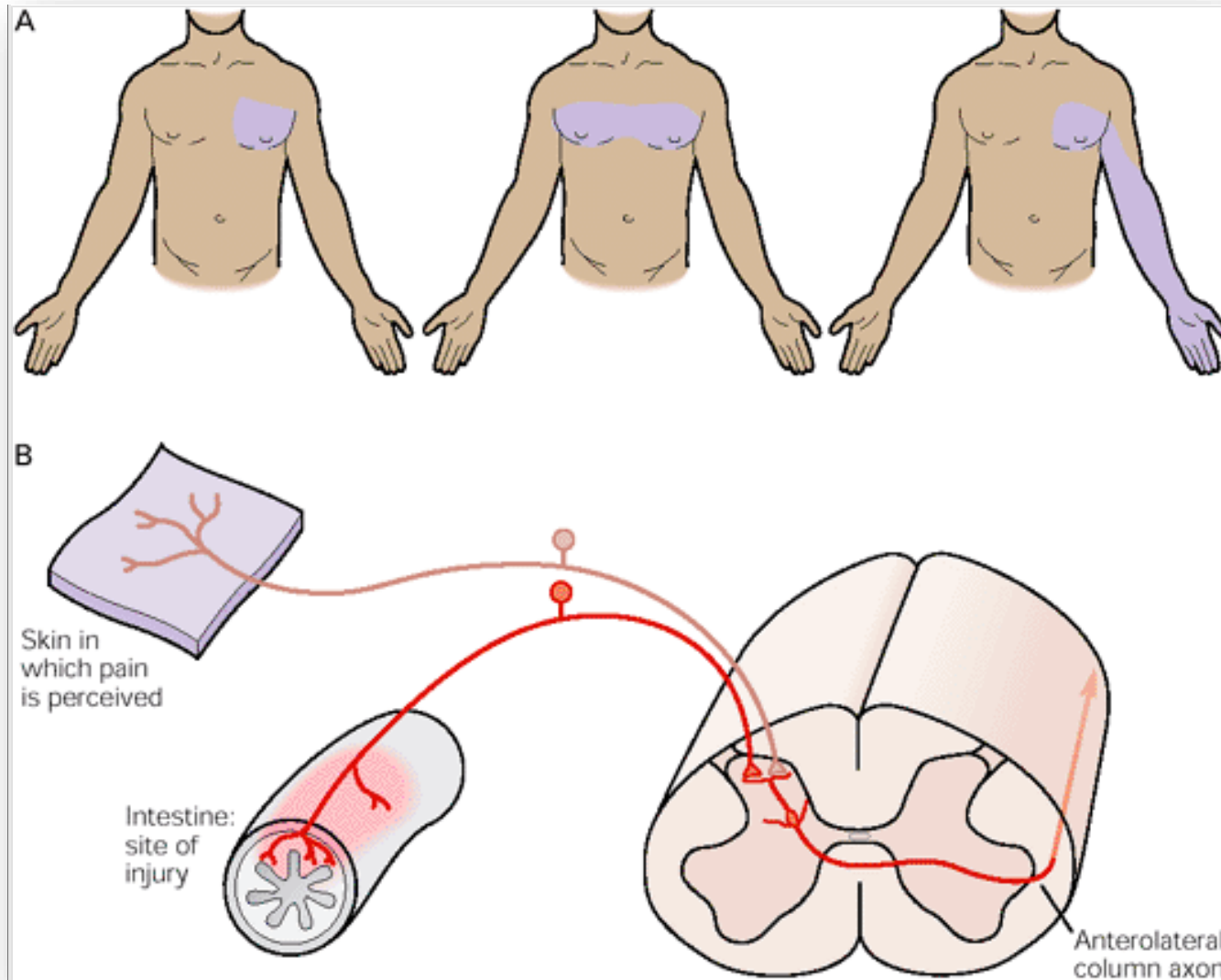
Five ascending pathways



Affective-motivational pain pathway

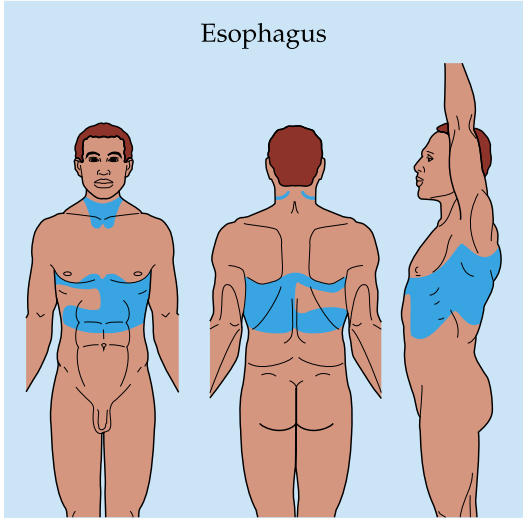


Referred pain

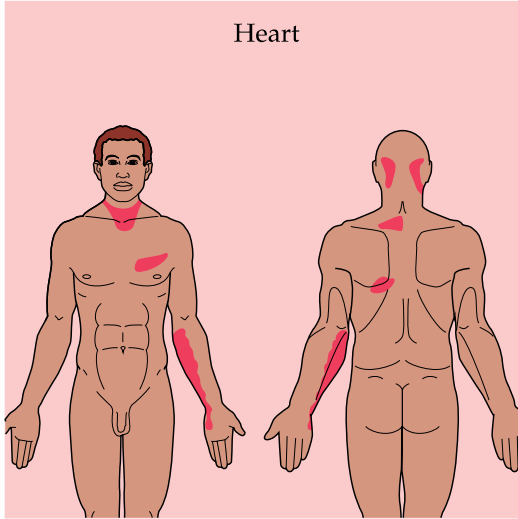


Referred pain

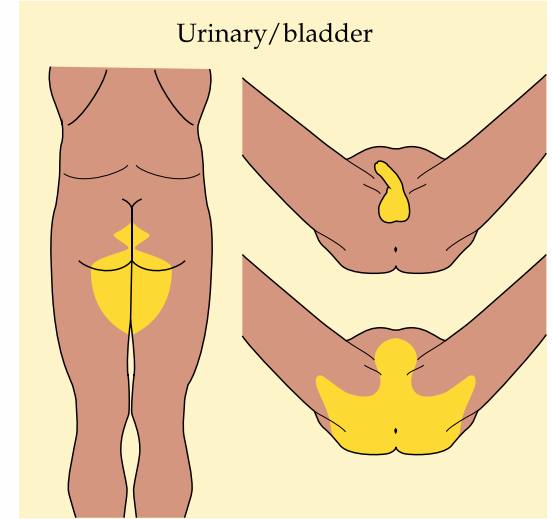
Esophagus



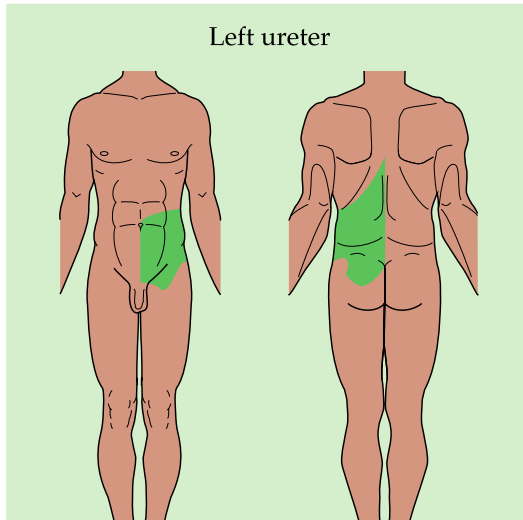
Heart



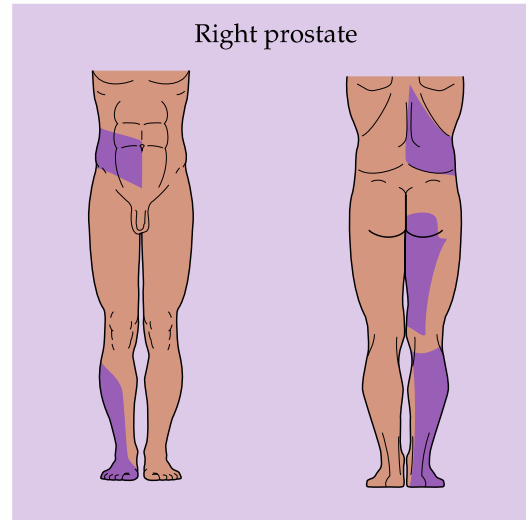
Urinary/bladder



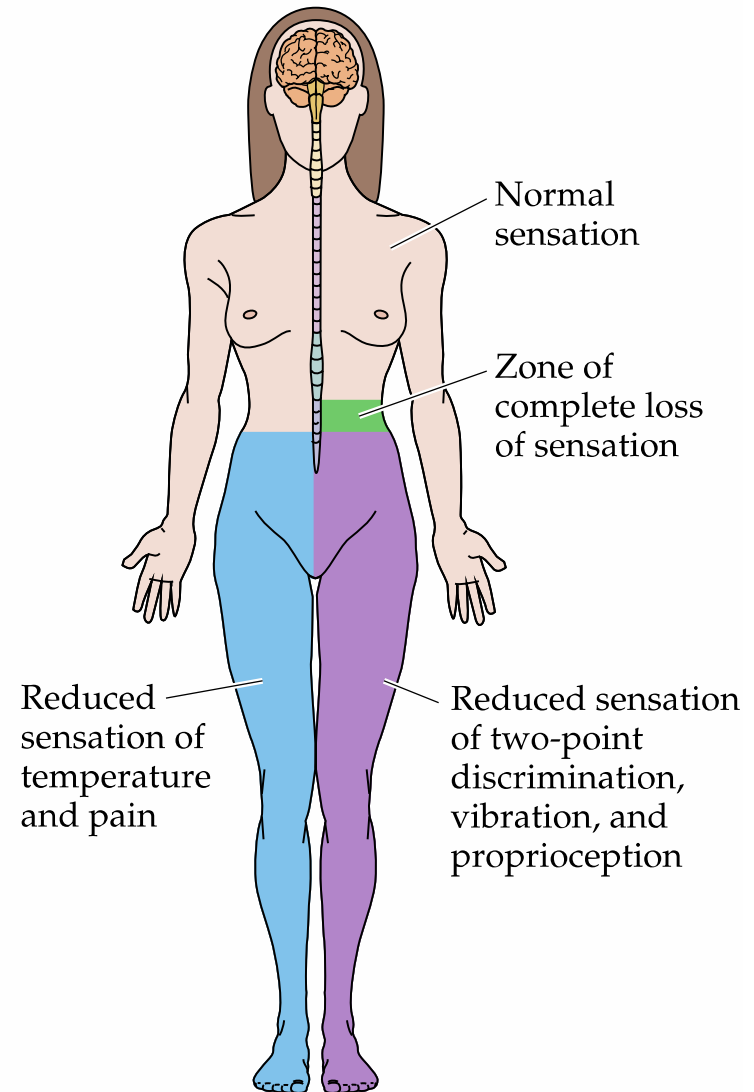
Left ureter



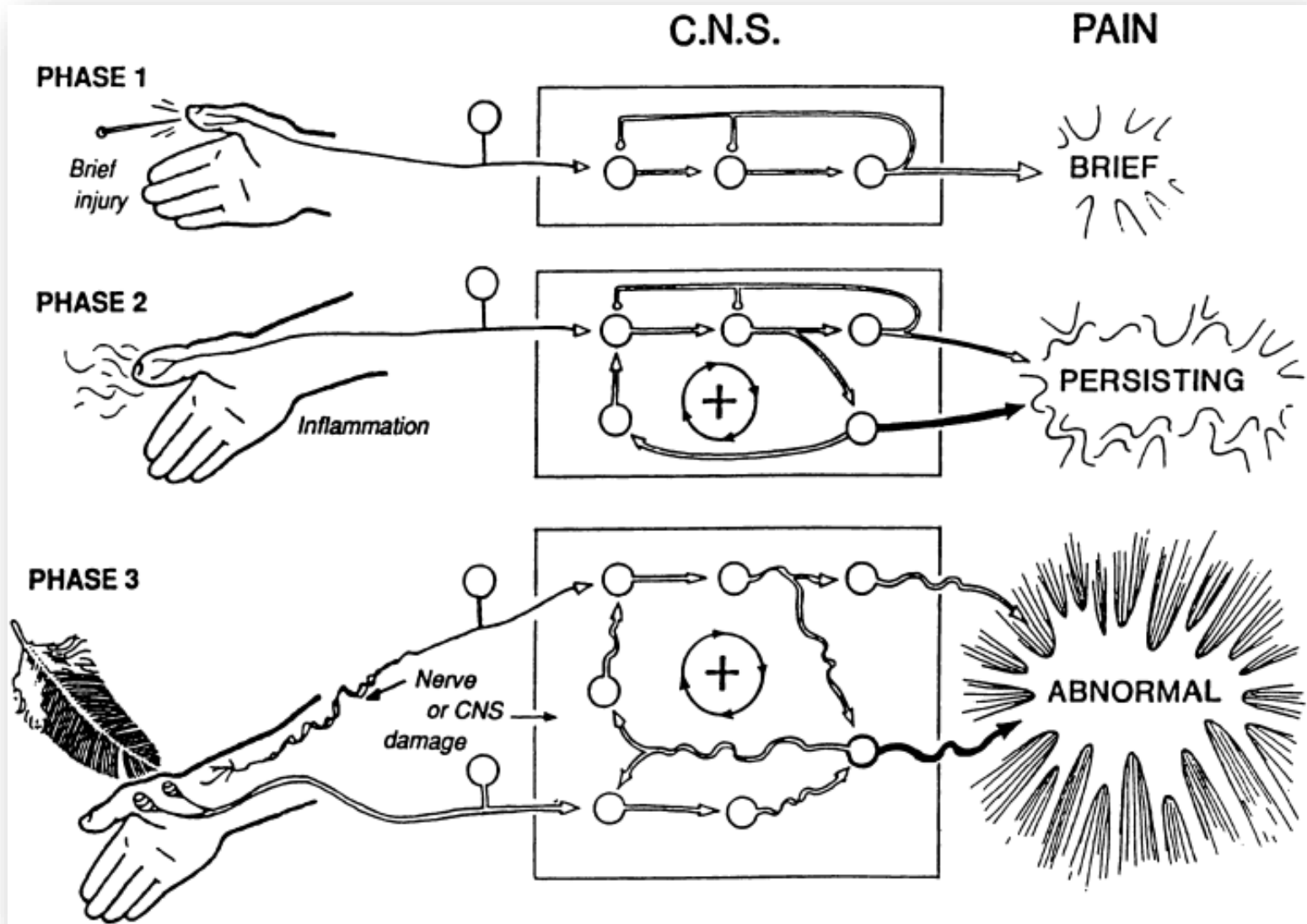
Right prostate



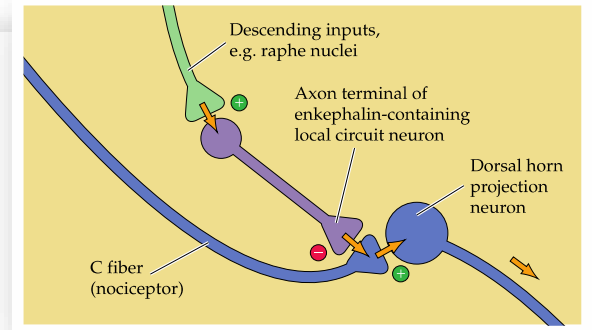
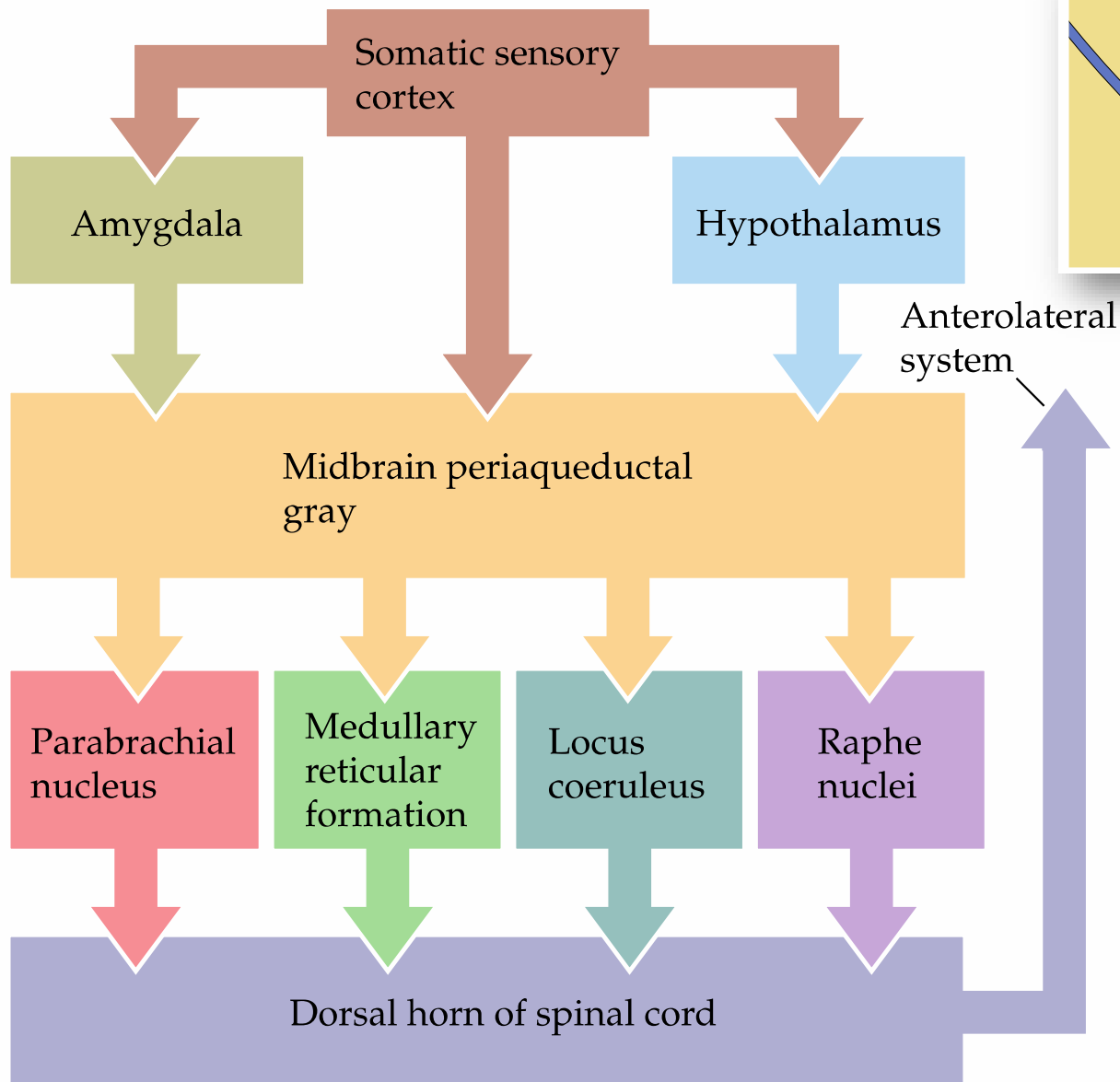
Dissociated sensory loss



Central and Peripheral Sensitization



Pain matrix and modulation

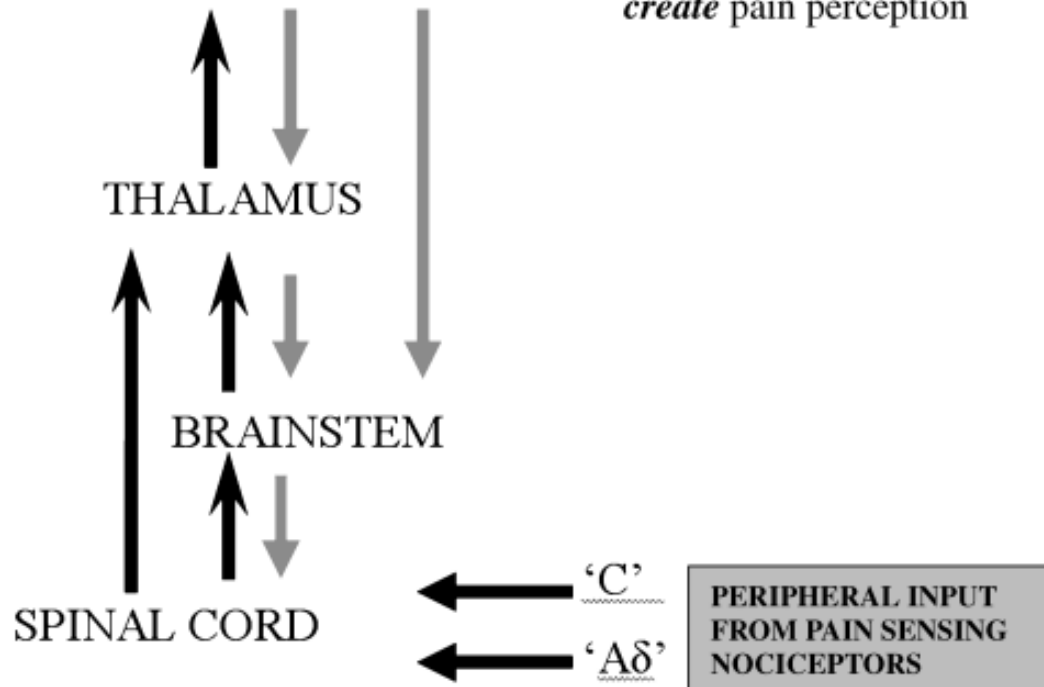


Pain matrix and modulation

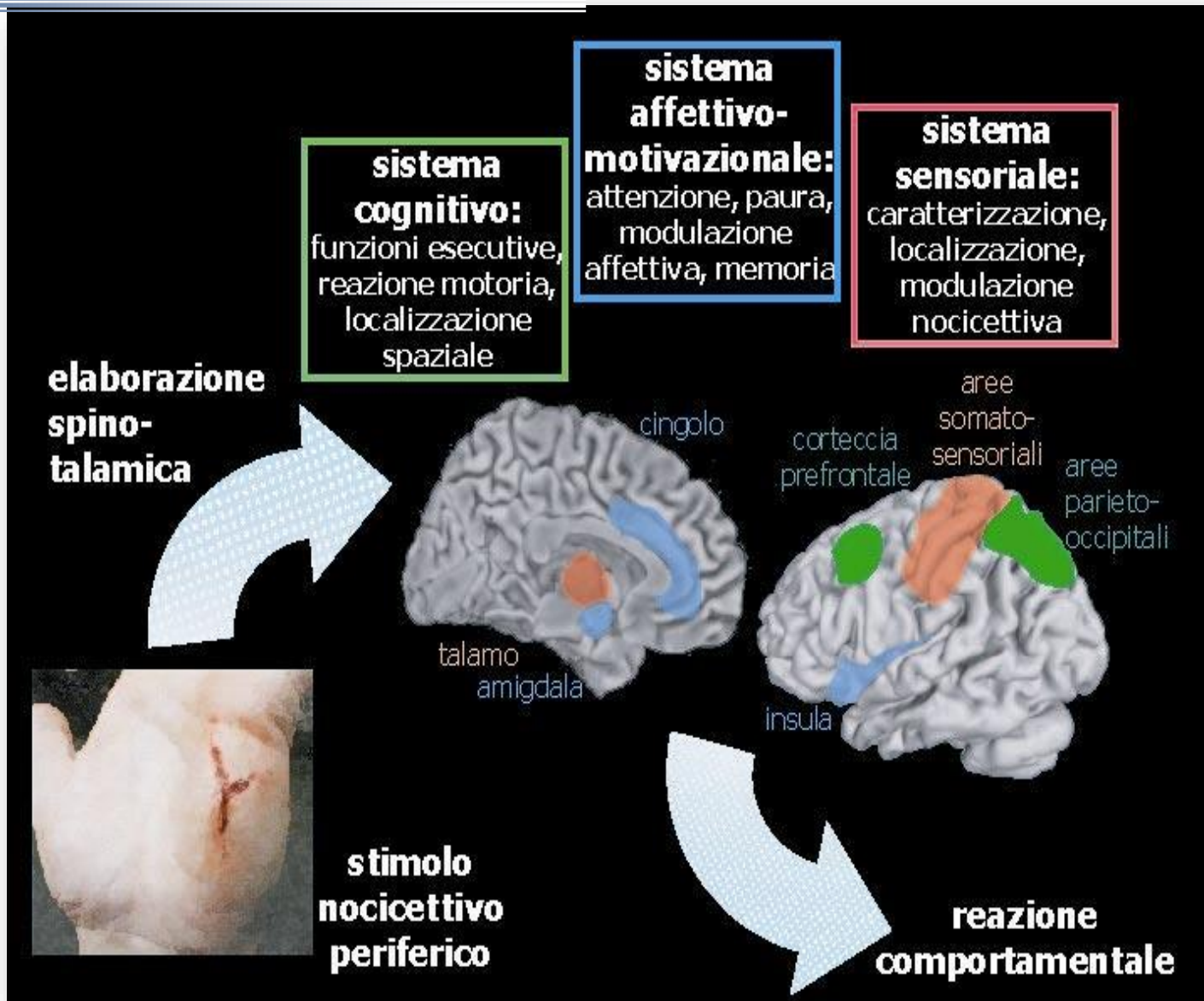
- INSULA
- SOMATOSENSORY CORTICES
- ANTERIOR, MIDDLE & POSTERIOR CINGULATE
- BASAL GANGLIA
- POSTERIOR PARIETAL CORTEX
- AMYGDALA
- HYPOTHALAMUS
- PREFRONTAL CORTEX

- Conscious perception of event being painful created by central *processsing* of incoming signals

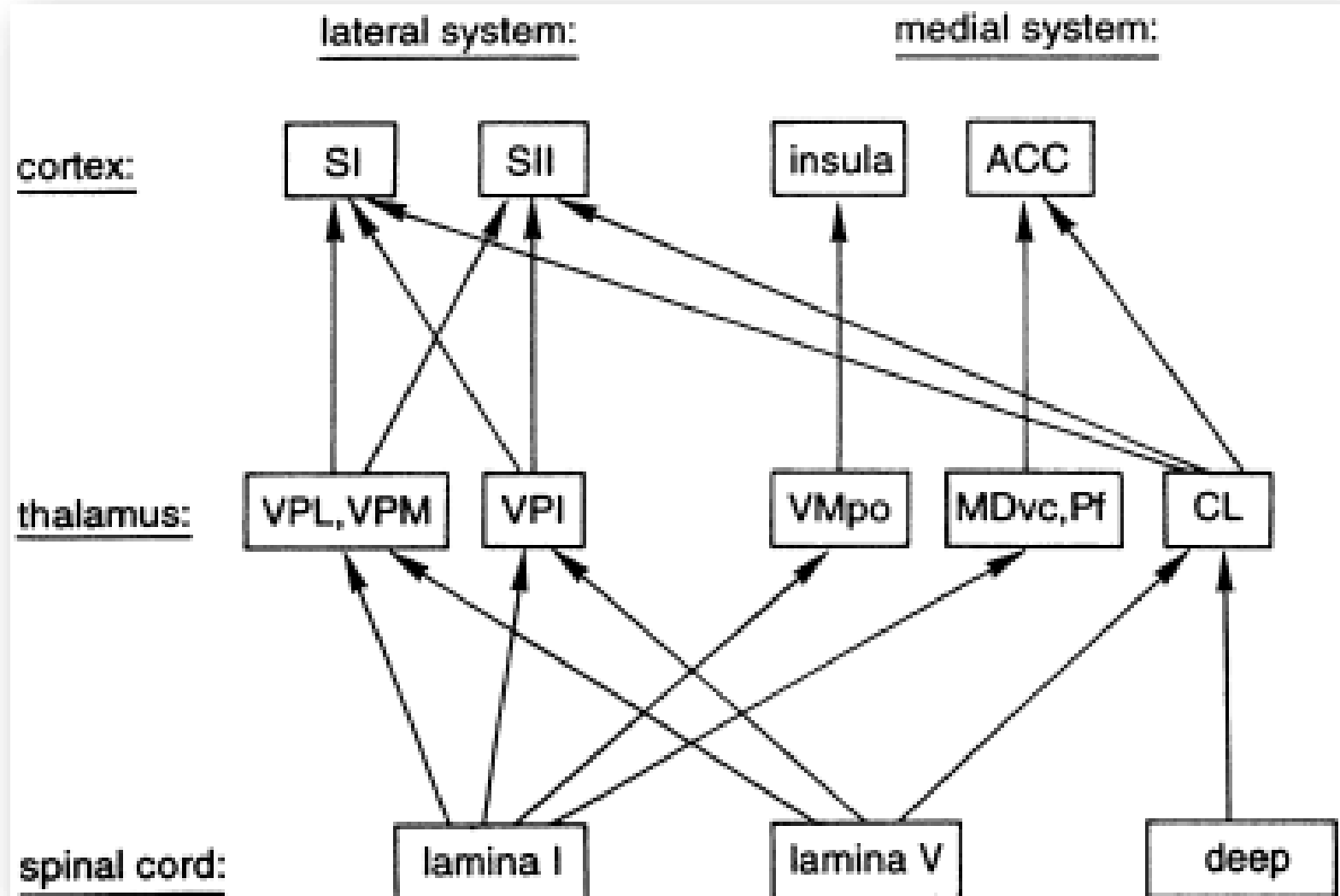
- Brain and spinal cord can *modulate*, but also *create* pain perception



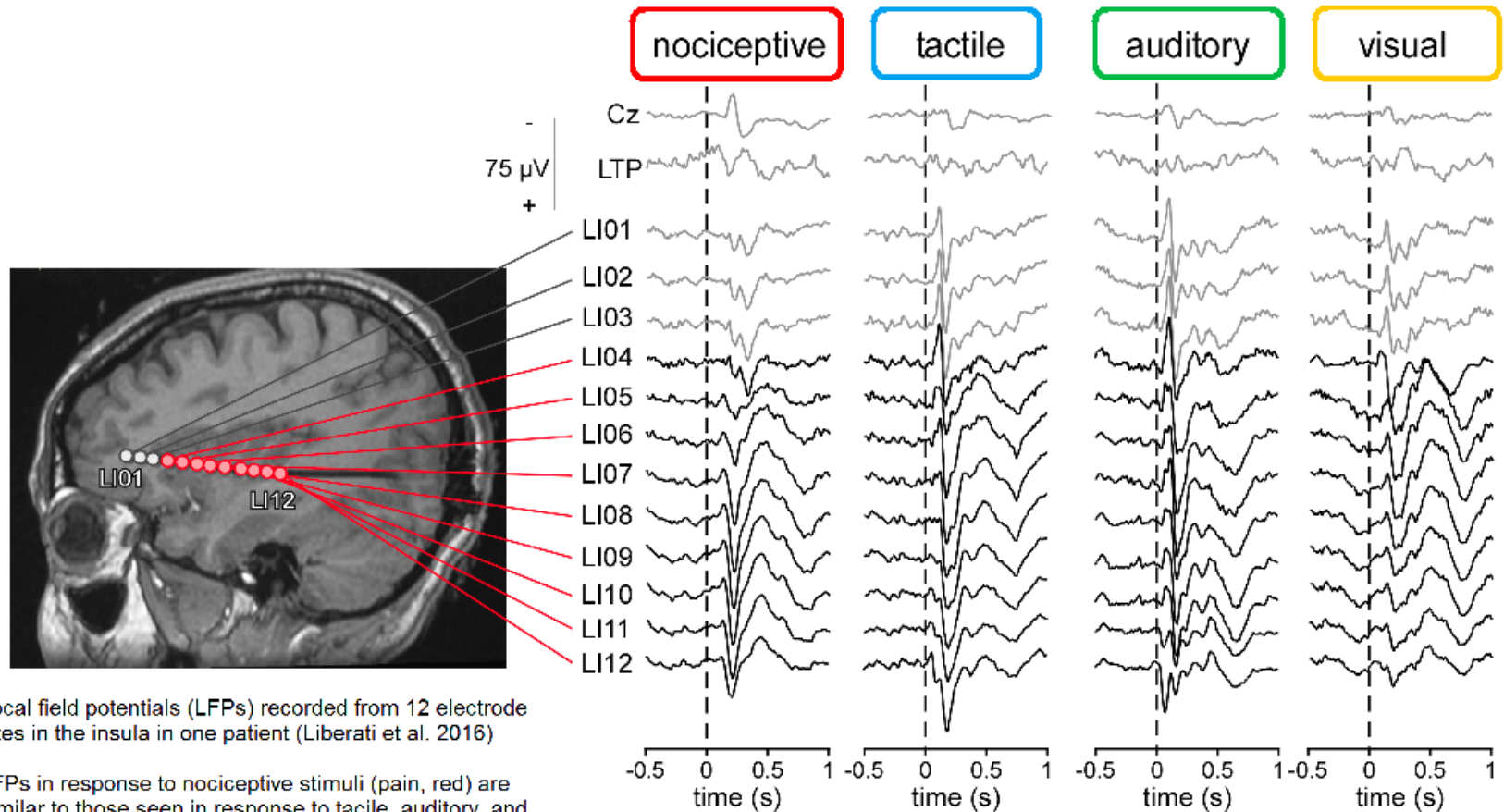
Pain neuromatrix



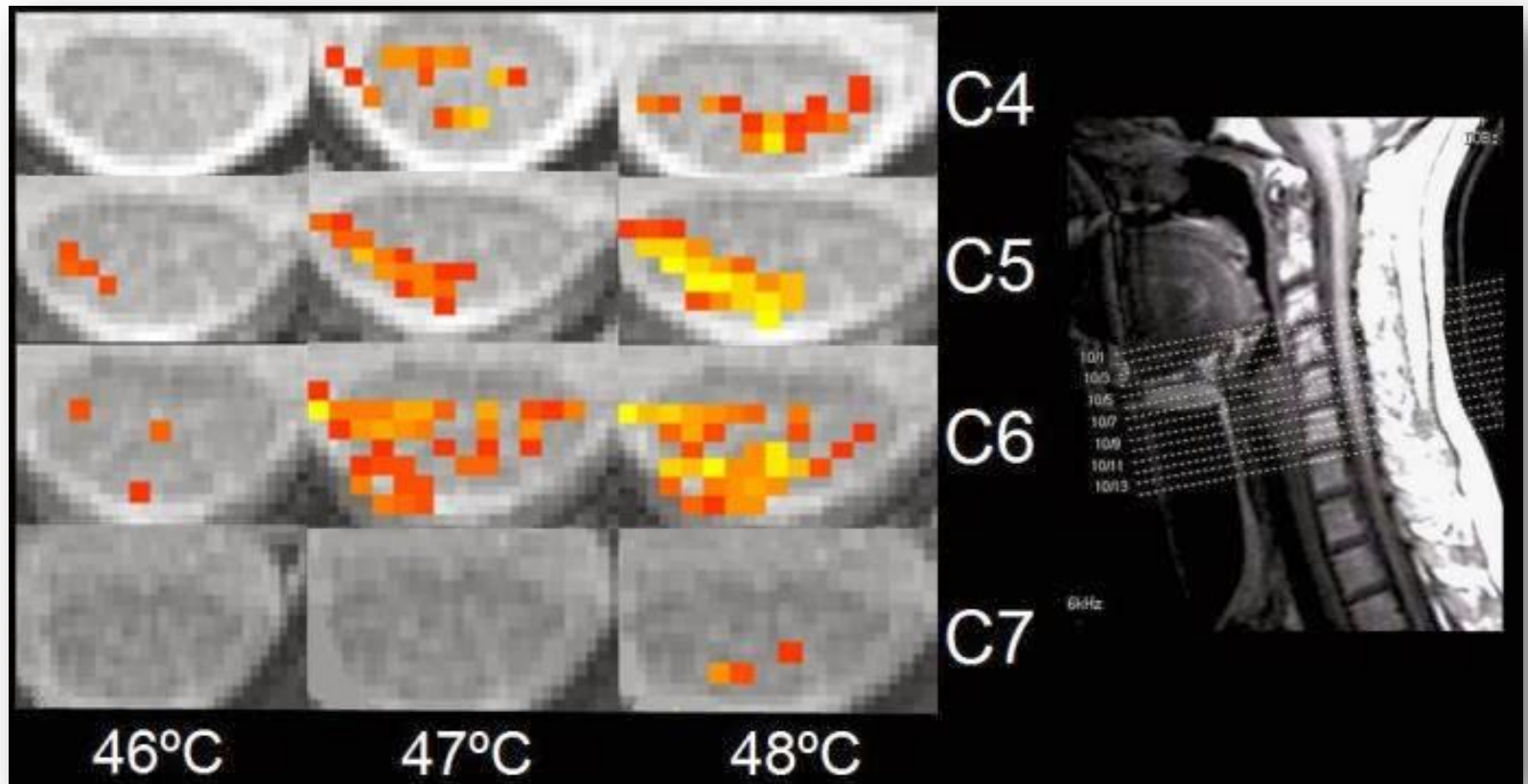
Pain neuromatrix



No primary pain cortex



Visualizing pain in the spinal cord



Anticipazione certa

Segnale



Stimolo

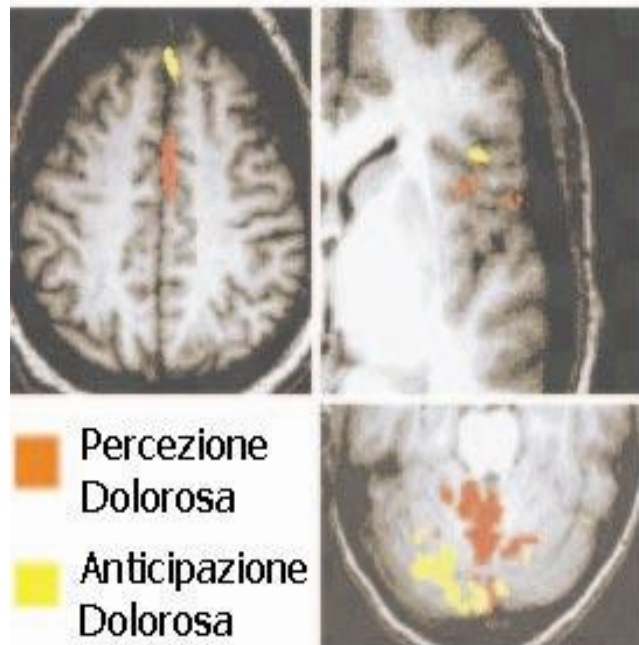
HT

Temp Alta
- Dolore



LT

Temp Bassa
- Non Dolore



Anticipazione dubbia

Segnale



Stimolo

HT

Temp Alta
- Dolore



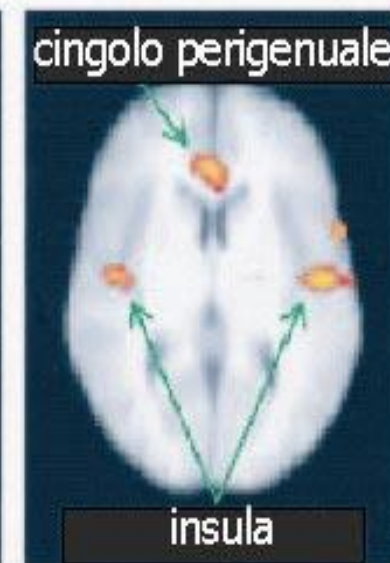
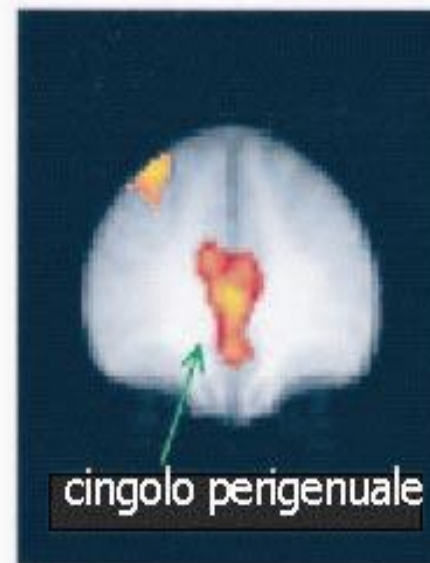
LT

Temp Bassa
- Non Dolore



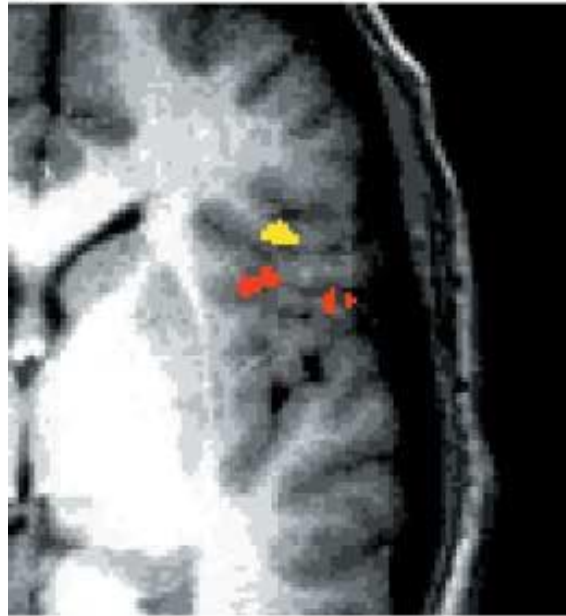
LT

Temp Bassa
- Non Dolore

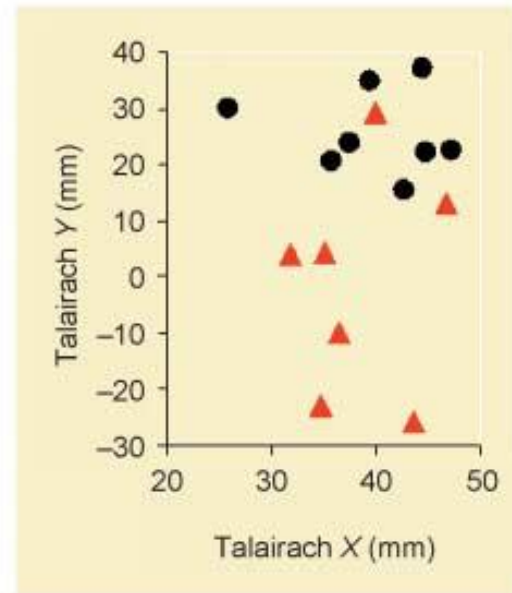


Anticipation of pain

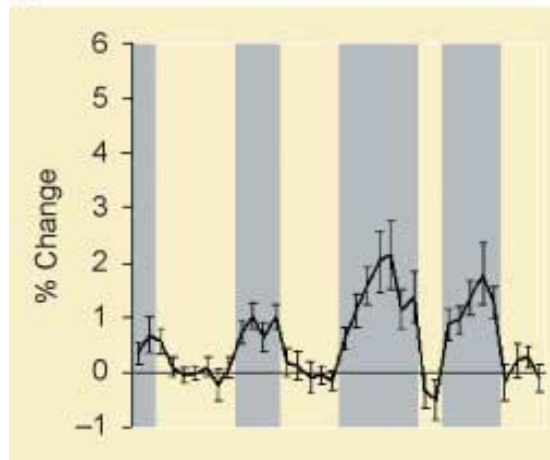
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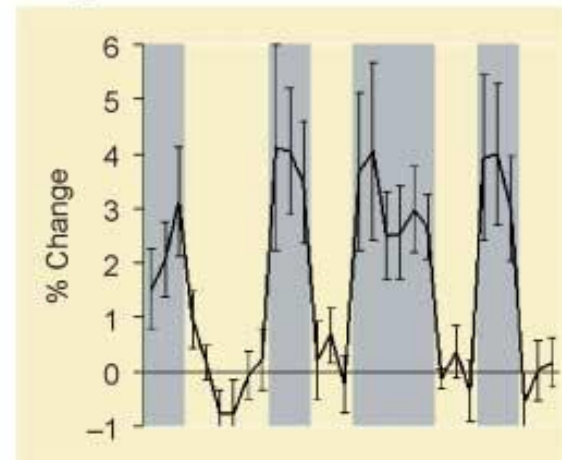
B



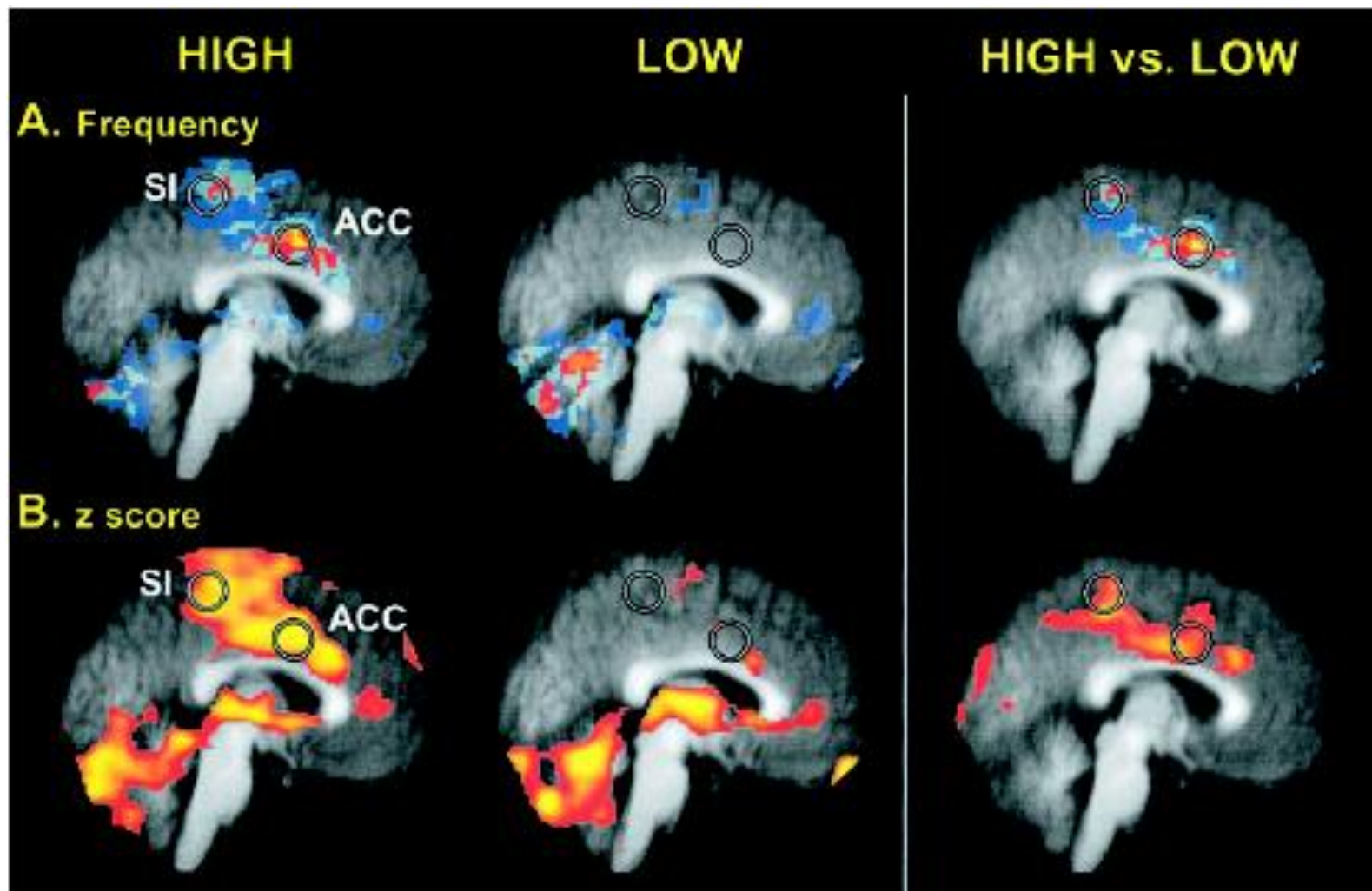
C



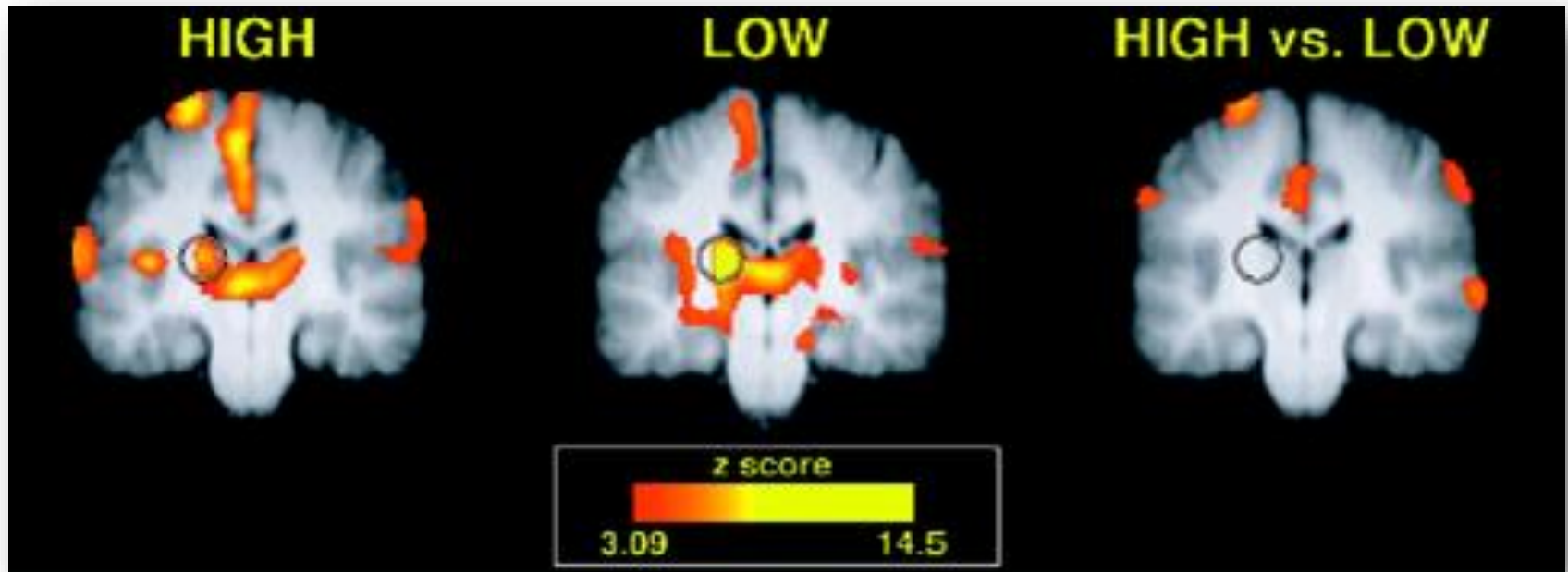
D



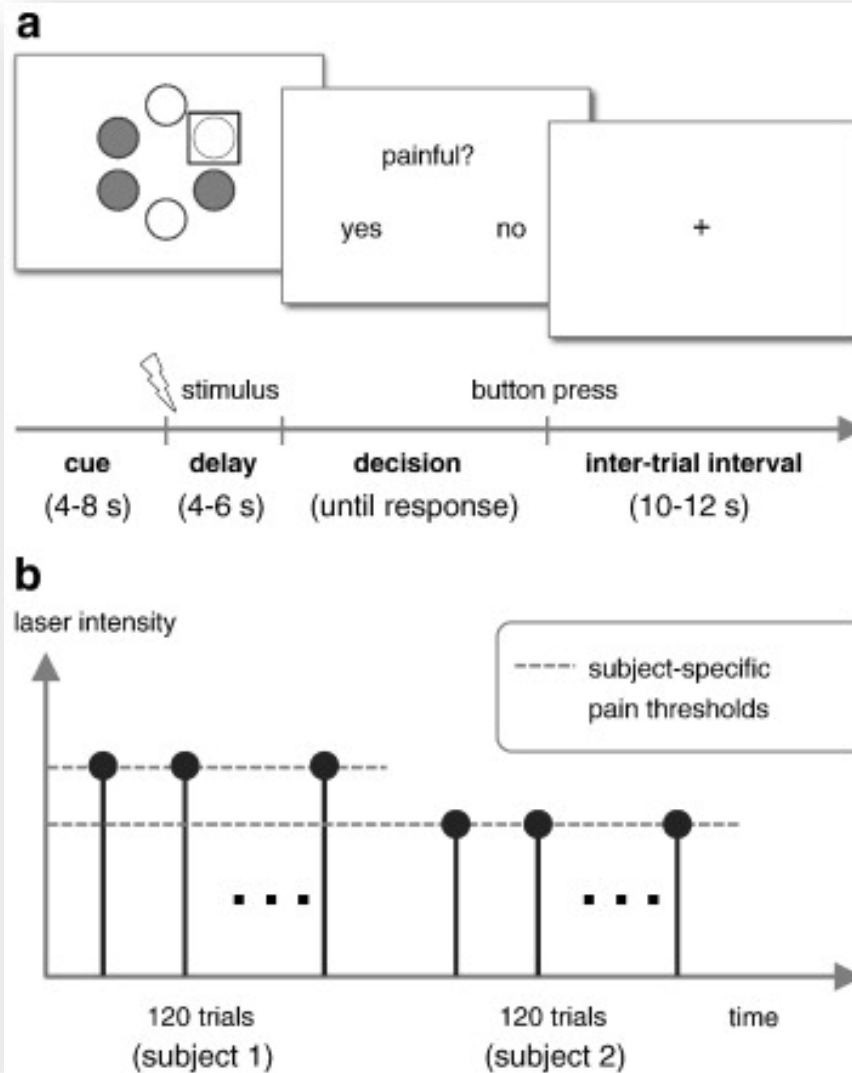
Sensitivity to pain



Sensitivity to pain

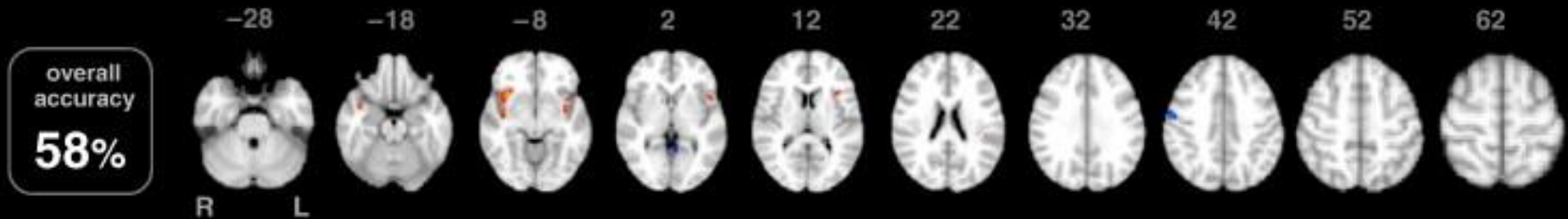


Pain perception and MVPA

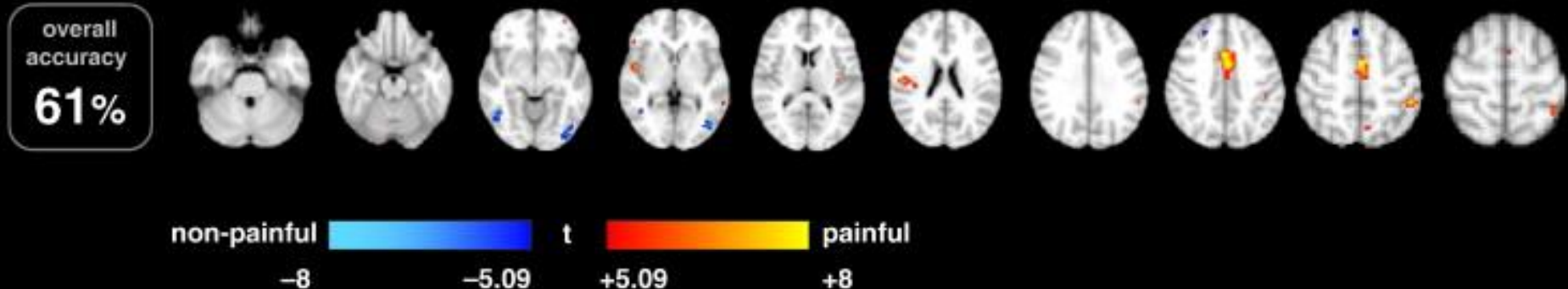


Pain perception and MVPA

a Decoding pain during anticipation ($p < 0.05$ FWE)

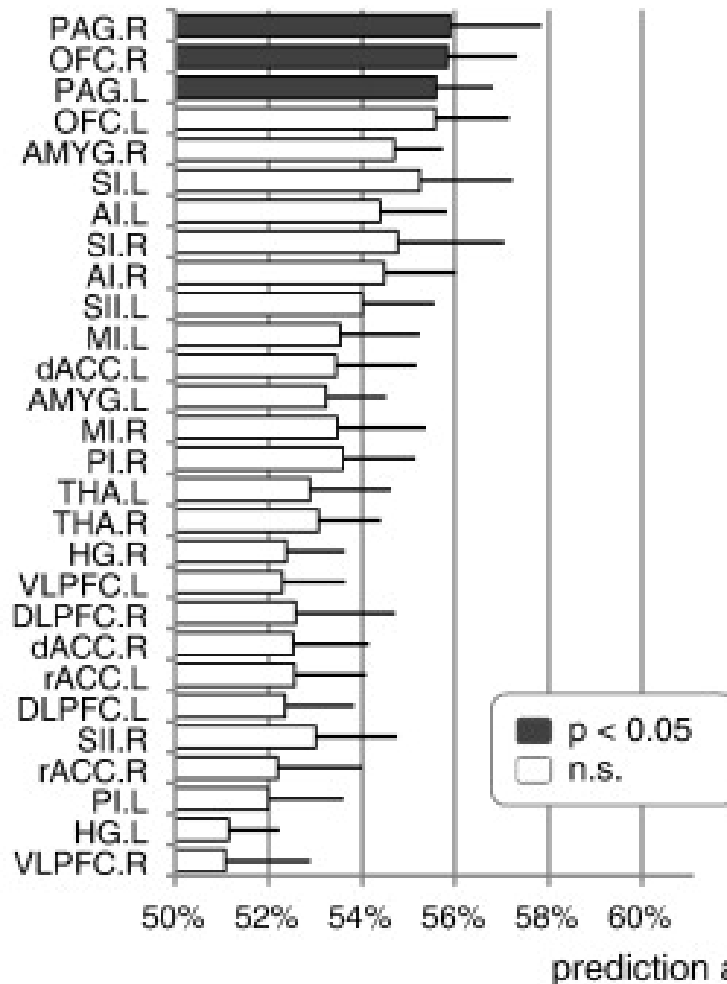


b Decoding pain during stimulation ($p < 0.05$ FWE)

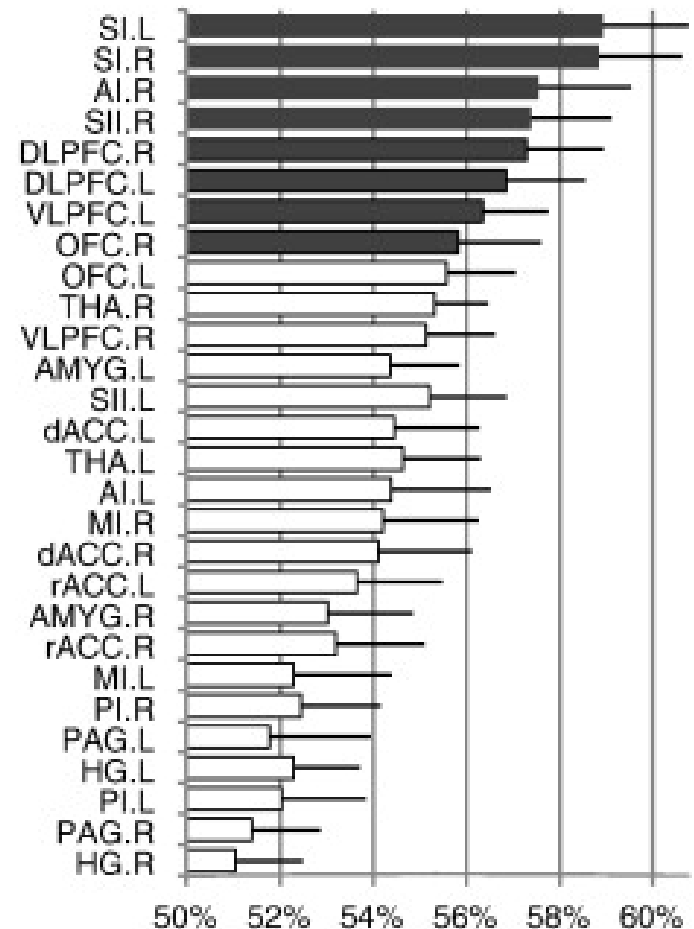


Pain perception and MVPA

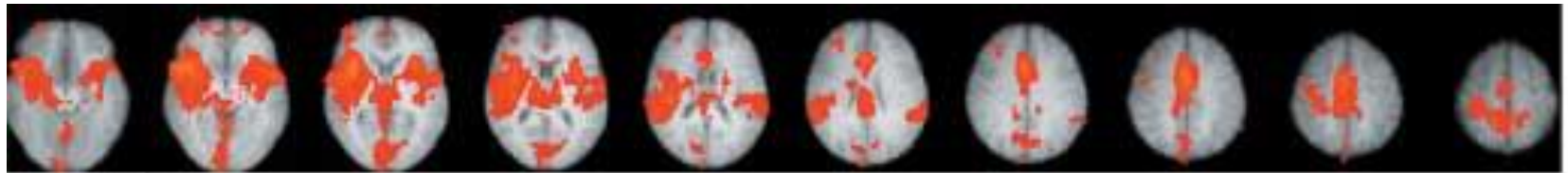
a Decoding pain during anticipation



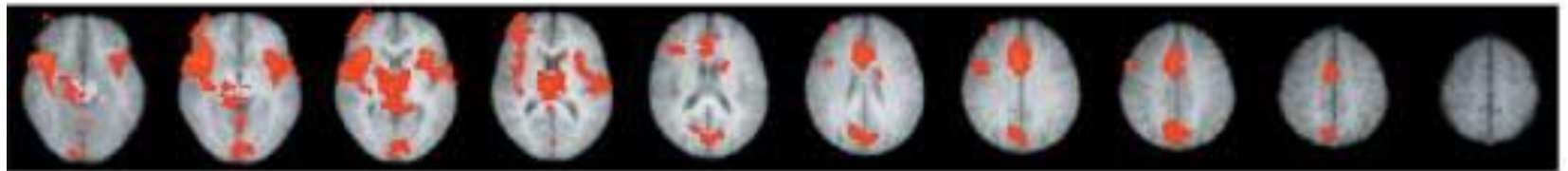
b Decoding pain during stimulation



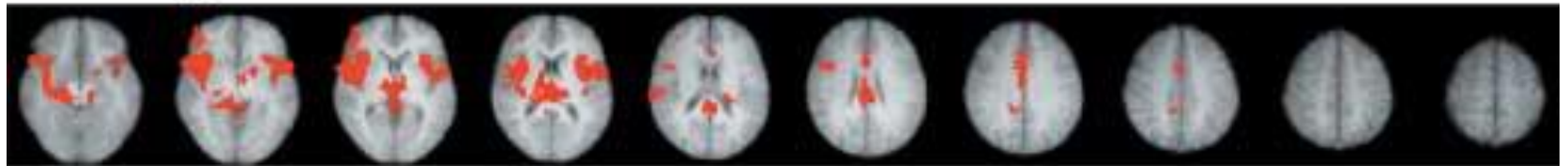
Pain perception and opioids



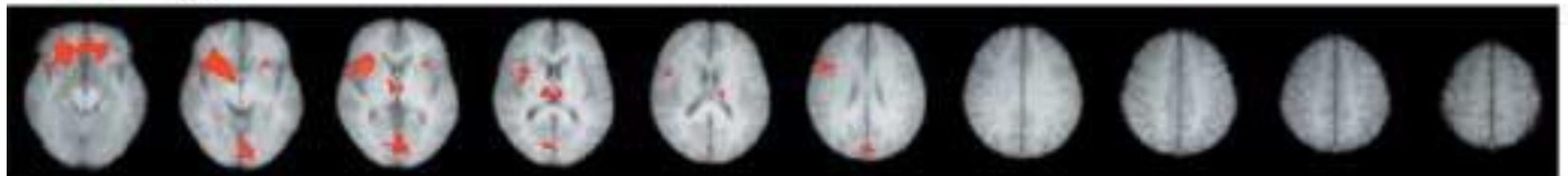
saline



0.5 ng, remifentanyl



1.0 ng, remifentanyl



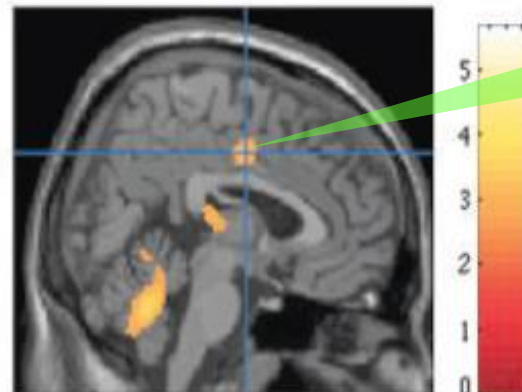
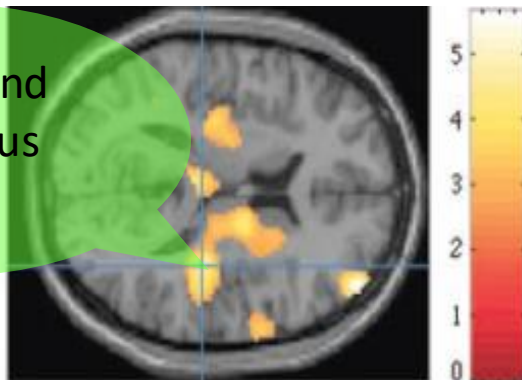
2.0 ng, remifentanyl

Placebo and Opioid Analgesia— Imaging a Shared Neuronal Network

Predrag Petrovic,¹ Eija Kalso,² Karl Magnus Petersson,¹ Martin Ingvar^{1*}
SCIENCE VOL 295 1 MARCH 2002

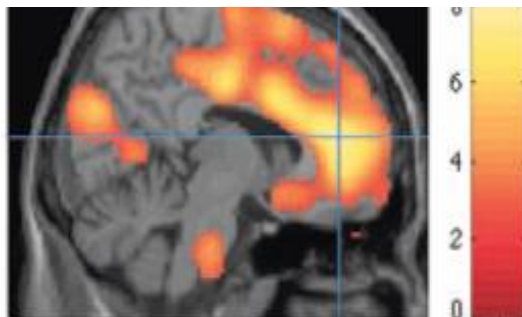
main effect of pain

Insula and
thalamus

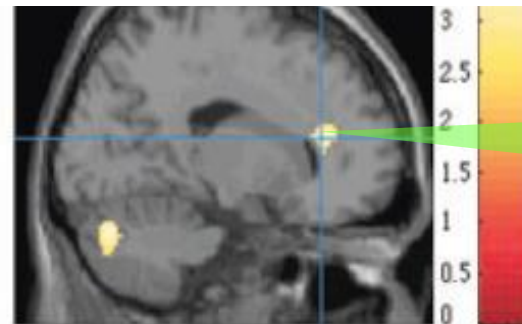


Caudal ACC

main effect of opioids

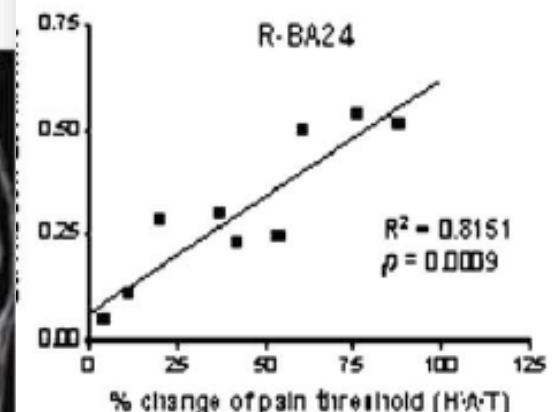
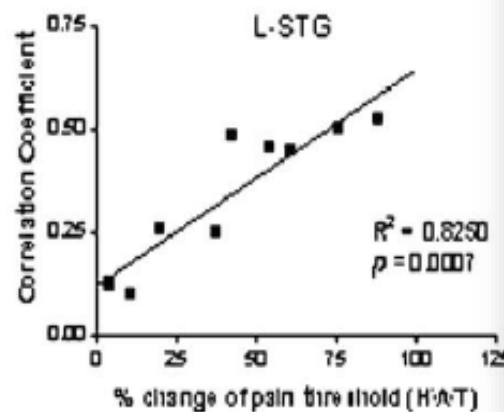
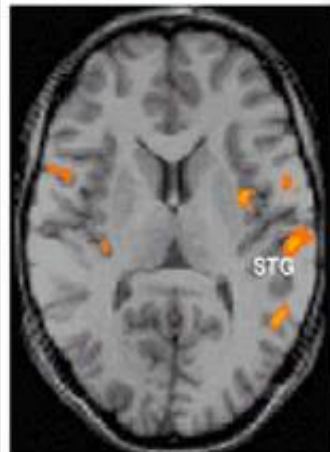
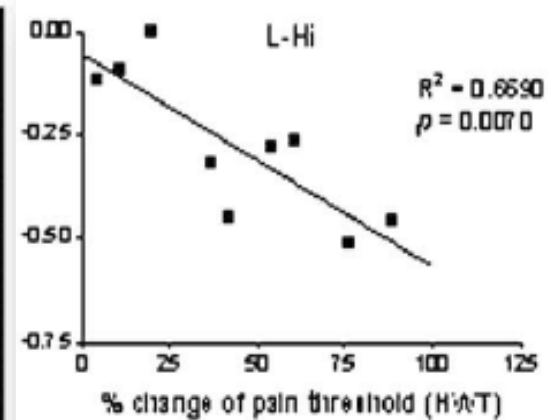
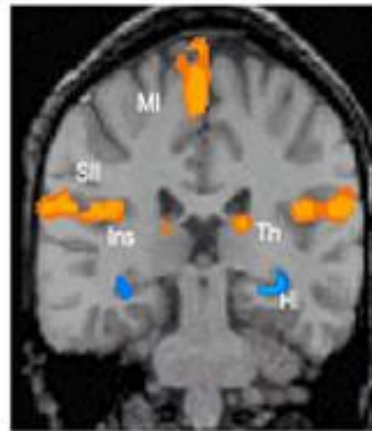
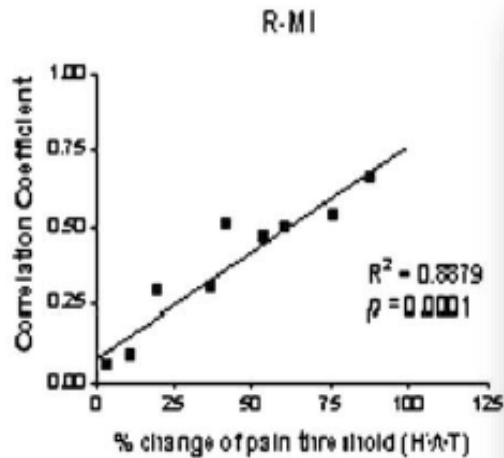
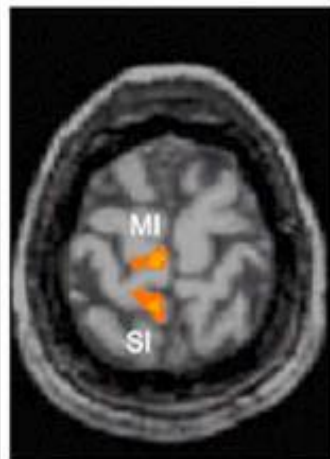


main effect of placebo

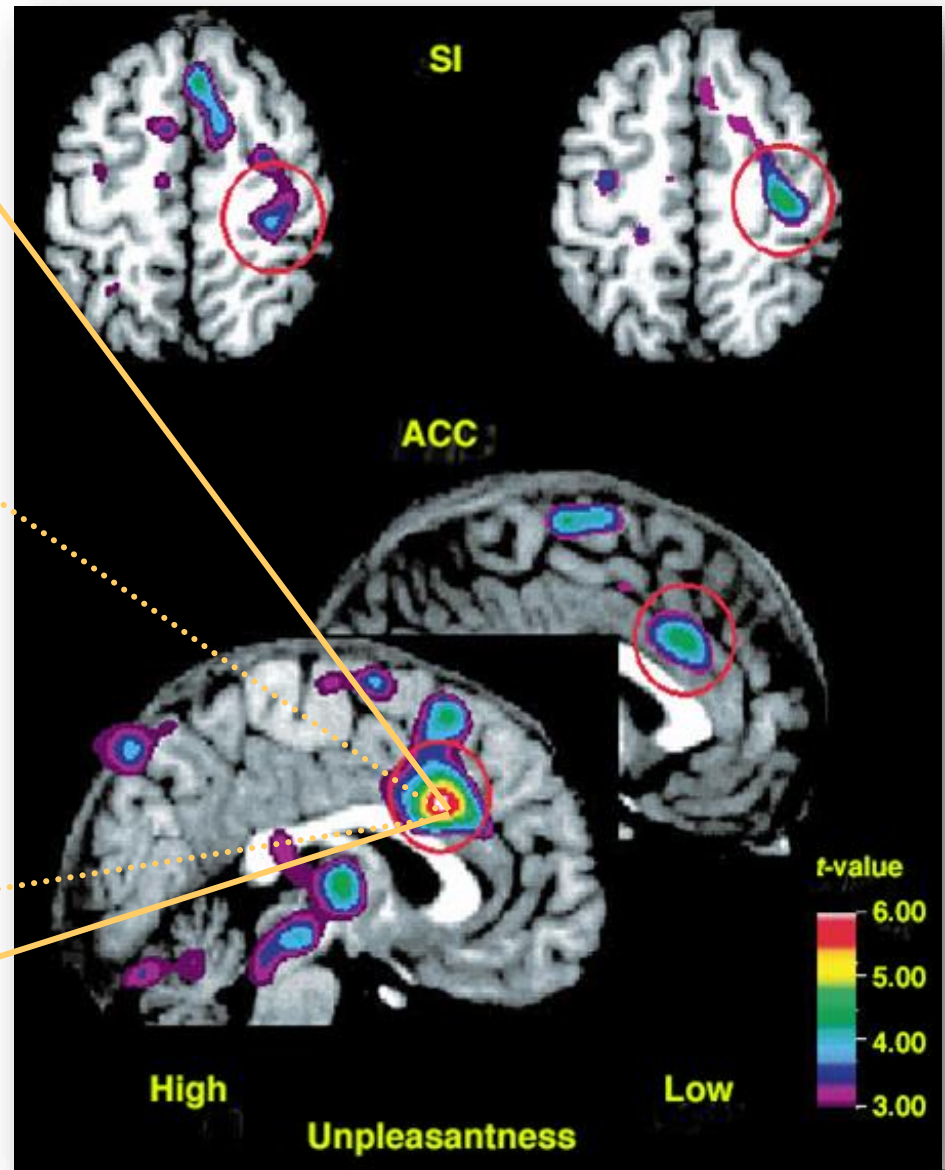
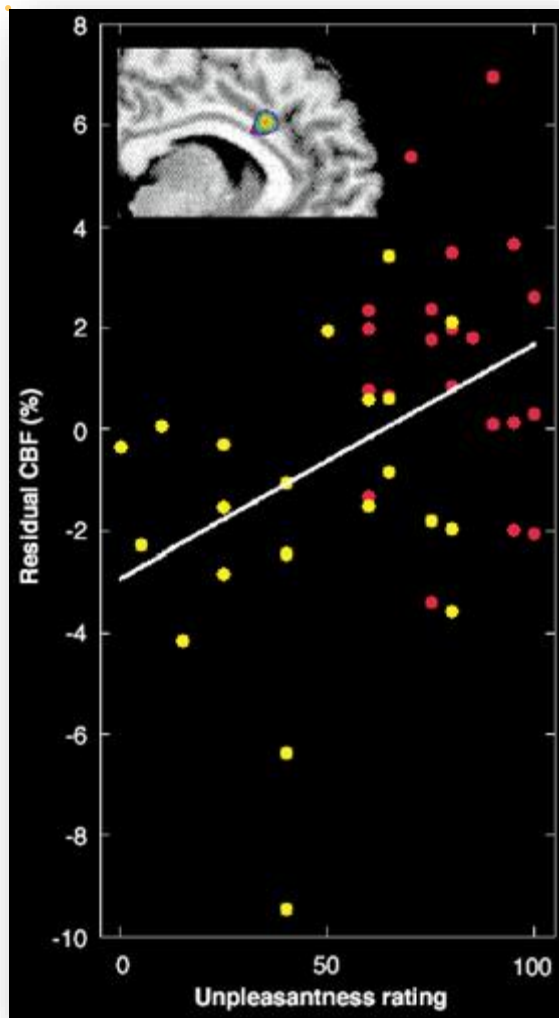


Rostral ACC

Relations between brain network activation and analgesic effect induced by low vs. high frequency electrical acupoint stimulation in different subjects: a functional magnetic resonance imaging study

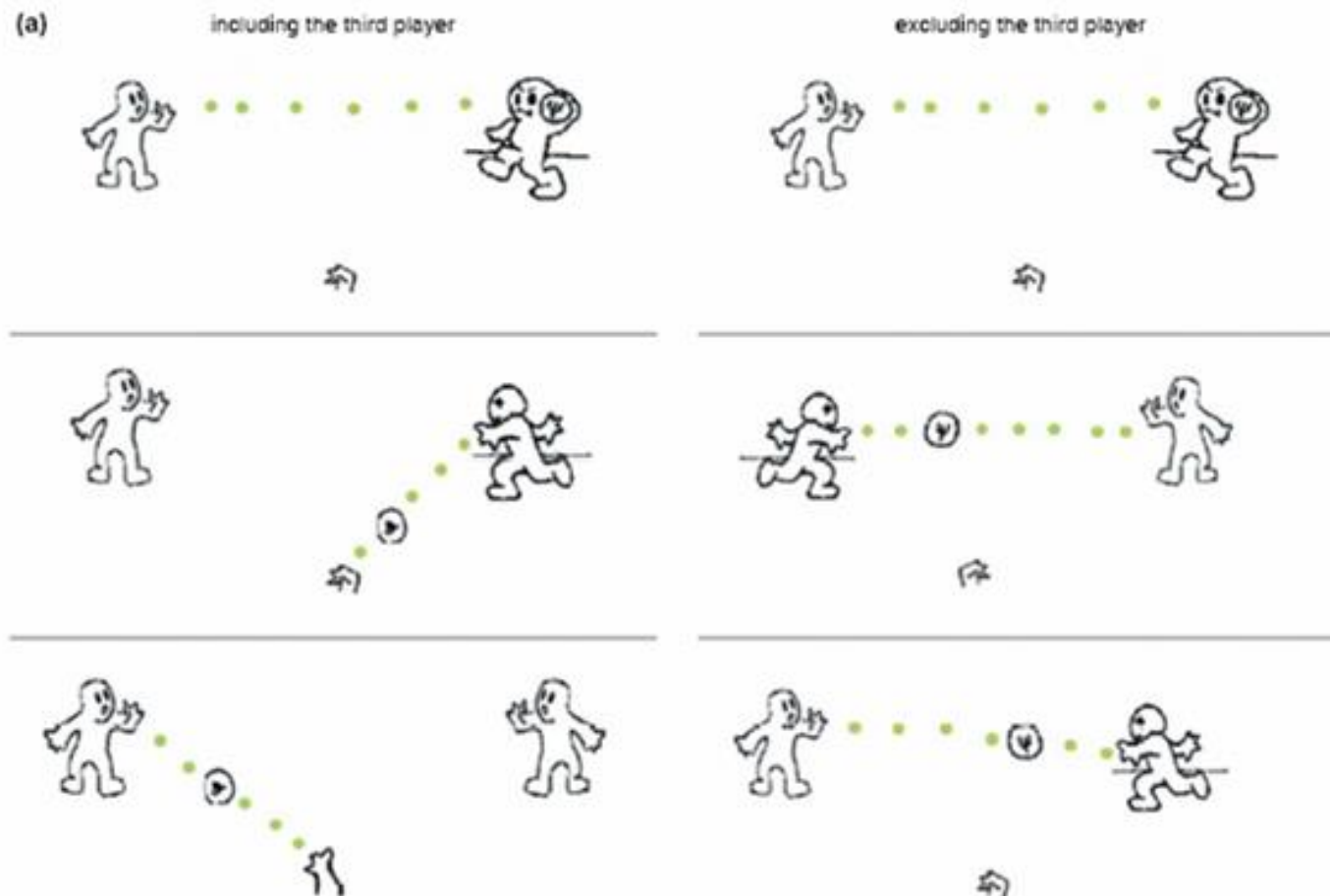


Modulation of pain



Does Rejection Hurt? An fMRI Study of Social Exclusion

Naomi I. Eisenberger,^{1*} Matthew D. Lieberman,¹
Kipling D. Williams²



Does Rejection Hurt?

Study of Social Exclusion

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